Algorithm 1 BinaryErrorLocating

Input: Solution set $S = \{y_i | i = 1, 2, ..., N\}$, mid step sampling num N_{mid}.
for $i = 1$ **to** N **do**
  $l \leftarrow 0$, $r \leftarrow$ StepLen(y_i) ▷ Define start and end index.
  while $l < r$ **do**
    $mid \leftarrow \lfloor (l + r)/2 \rfloor$
    MID_RES $\leftarrow \mathcal{F}(\{y_{i,1}, y_{i,2}, ..., y_{i,mid}\}, N_{mid})$ ▷ Sampling from mid step.
    if VERIFY(MID_RES) does not contain True **then** ▷ Judge sampling's final answer.
      $r \leftarrow mid$
    else
      $l \leftarrow mid + 1$
    end if
  end while
  $e_i \leftarrow$ WRONGSTEPLABELING(y_i, l) ▷ Collect sequential labels of y_i.
  ErrorLocatingSet.append$\{e_i\}$
end for
Output: ErrorLocatingSet

Algorithm 2 PS-GRPO

Input: Policy model π_θ, Process Reward Model ϕ_θ, train dataset $D_{RL} = \{I_i, Q_i, Y_i\}_{i=1}^{N}$.
ErrorLocatingSet $\leftarrow$ []
for Epoch $= 1$ to N **do**
  for Batched data $\{I, Q, Y\}$ in D_{RL} **do**
    Generate rollouts of $\{I, Q\}$: $\{c^j\}_{j=1}^{M} \sim \pi_\theta$
    Calculate process reward sets $S = \{(p_1, p_2, \cdots, p_{|c_j|}) \sim \phi_\theta(c^j) | 0 < j < M\}$
    Calculate reward $\{r^j\}_{i=1}^{M}$ by Equation 5 and 6.
    Calculate relative advantages $\{\hat{A}^j\}_{i=1}^{M}$ using Equation 7.
    Update policy model π_θ using Equation 8.
  end for
end for
Output: Updated policy model π_θ.

G Prompt Design

G.1 Prompt Utilized in MMathCoT-1M Synthesis

In this section, we provide the specific prompts for three-module data synthesis. Additionally, Gemini-1.5-Flash is a model that is very sensitive to prompts and parameters in practical experience, and we will share detailed adjustment experiences.

CoT Expansion CoT expansion prompt for answer-only data source can be seen in Figure 12. We order the Gemini-1.5-flash to give a reasonable process directing to the ground-truth. After the execution, we find that the outputs is not so clear. The model sometimes will give trajectories that include "we must trust the answer" or "let me assume". We identify these phrases as signals that the model can not solve the problem naturally and independently. We will filter these samples.

Analysis Rewriting Rewriting prompt for analysis-formatted data synthesis is illustrated in Figure 13. For solutions in an analytical format, we transform them into clear step-by-step format trajectories. During this process, Gemini-1.5-flash-002 does not exhibit significant questioning or make conditional requests. We improve data quality through reorganization and polishing of the language logic.

Format Unified By employing a unified format prompt shown in Figure 14 to modify the reasoning styles of plan-and-reasoning and symbolic approaches, we are able to extract a more natural language process aligned with the pre-training style. A single example is sufficient to elicit perceptually favorable responses.

CoT Distillation

For a question, you need to provide a **step-by-step solution** that directs to the given right answer. You must **trust** the given correct answer and avoid **making any requests** for additional information and **any suspect**.

Example 1:

Question: Given that points A, B, and C lie on circle O and angle AOB measures 80.0 degrees, what is the measure of angle ACB? Choices:

A: 80°

B: 70°

C: 60°

D: 40°

Provided answer: D

Your Response:

Step 1: Identify the given information: angle AOB = 80°.

Step 2: Apply the relevant theorem: angle ACB = 1/2 angle AOB (inscribed angle theorem).

Step 3: Substitute the given value: angle ACB = 1/2 * 80° = 40°.

Step 4: State the answer: The answer is D.

†Answer: D

Example 2:

Question: As shown in the diagram, AB is the diameter of circle O. EF and EB are chords of circle O. Connecting OF, if angle AOF = 40° , then the degree of angle E is (). Choices:

A: 40°

B: 50°

C: 55°

D: 70°

Provided answer: D

Your Response:

Step 1: Given angle AOF = 40°.

Step 2: Angles AOF and FOB are supplementary, meaning they add up to 180°. Therefore, angle FOB = 180° - 40° = 140°.

Step 3: Angle E is half of angle FOB. Therefore, angle E = 0.5 * 140° = 70°.

Step 4: The answer is D.

†Answer: D

Now, let's begin.

Question: {QuestionText}

Provided answer: {SolutionText}

Your Response:

Figure 12: CoT expansion prompt for answer-only data.

Double-checking After completing the above three points, we apply an LLM-as-a-judge for double-checking the synthesized data, ensuring that the solutions do not contain unreasonable processes, such as untimely questioning, conditional requests, or reasoning loops. The specific prompt design is shown in Figure 15. After this layer of filtering, we obtain the final MMathCoT-1M.

G.2 Prompt Utilized in DualMath-1.1M Synthesis

In this section, we demonstrate the prompts used in MIE.

- **Geometry Problem**: For geometry problem, we prompt the Gemini-1.5-Flash-002 to first identify key geometry features in the figure. We then order it to introduce a misinterpretation on these elements. Finally, use the wrong information to execute a misleading solution. The total design can be seen in Figure 16.

Analysis Rewriting

I have a mathematical problem-solving process here, and both the process and the final answer are completely correct. I hope you can solve it in a **different way** based on the existing solution process, ensuring that the process and results are correct, especially the reasoning results **remain consistent with the original**.
Please remember the following instructions:

1. You must **trust** that the solution and final answer is correct, and you **do not need to review or refute it**.
2. Your transcription must be **semantically coherent**.
3. You **cannot make requests** like "require more information" or similar, because the given conditions are sufficient and the solution is definitely correct.
4. You can rewrite from multiple perspectives, such as **providing different solutions** to the problem or adopting **different textual styles**.
5. You must end with '†Answer: ', filling in the definitive answer provided.

Question:
{QuestionText}
Correct Solution:
{SolutionText}
Give the rewrited solution in the following format:
Step 1: ...
Step 2: ...
...
†Answer: ...

Your solution:

Figure 13: Analysis rewriting prompt for analysis-formatted data.

- **Charts & Function**: For ChartQA and math functions, we prompt Gemini-1.5-Flash-002 to first check the fine-grained data points. We then attempt to insert spatially similar data to induce a misinterpretation. This subsequently leads to incorrect solutions for automatic labeling.
- **LLM-as-a-Judge**: For chart reasoning and function problem, we execute similar process on Gemini-1.5-Flash-002. We place it in Figure 17.

H Case Study

H.1 Showcase on Best-of-N Evaluation

To more clearly illustrate the effectiveness of URSA-8B-RM in BoN evaluation, a case on MathVista-GPS is demonstrated (Figure 18). This case shows that URSA-8B-RM is sensitive to false theorem application and misunderstandings of angle-number relations. The good property not only enables URSA-8B-RM to perform well in BoN evaluation but also endows it with the potential to identify more valuable learning samples in online reinforcement learning.

H.2 Process of Misinterpretation Insertion Engine

As shown in Figure 19, MIE performs three main actions: First, it interprets the mathematical information in the image. Then, it replaces key information at a selected step. Finally, it continues reasoning based on the modified conditions.

H.3 Failure Pattern in Process During GRPO

In this section, we intuitively reveal through examples why PS-GRPO works effectively. We first introduce the concept of false-positive rollouts, which are rollouts that, despite reaching the correct answer, do not provide perfect intermediate actions to arrive at the solution. They can generally be